
SAFESTRESS-ML: TRUSTWORTHY MULTIMODAL STRESS DETECTION VIA TREE-BASED ENSEMBLES AND EXPLAINABLE AI

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ABSTRACT

Most multimodal stress detection systems rely on deep neural networks (DNNs), which are computationally heavy for wearable devices and lack clinical transparency. I present SafeStress-ML, a lightweight pipeline based on tree-based ensembles, selecting XGBoost as the primary classifier, that avoids deep learning architectures entirely. My pipeline combines subject-wise Z-score normalization, targeted statistical feature extraction from electrodermal activity (EDA) and blood volume pulse (BVP), and in-fold SMOTE balancing. Experimental trials evaluating window sizes (30s vs. 60s), sampling strategies, and probability thresholding were conducted. Evaluated on the public WESAD dataset using 15-fold Leave-One-Subject-Out (LOSO) cross-validation, my method achieves a reliable average accuracy of 89.9% and a Macro F1-score of 86.26%. Furthermore, integrating TreeSHAP provides clear, feature-level explanations for individual predictions, identifying EDA as the primary indicator of sympathetic response. This approach balances computational efficiency with the transparency requirements of modern healthcare frameworks and the EU AI Act.

1 Introduction

Chronic psychological stress contributes significantly to cardiovascular diseases, immune system suppression, and mental health disorders. For preventive medicine, objective and continuous monitoring of stress in real-world settings is highly valuable. While traditional stress assessment relied on subjective questionnaires, consumer wearable devices now enable non-invasive tracking of the Autonomic Nervous System (ANS). Specifically, sympathetic activation can be monitored via Electrodermal Activity (EDA), reflecting sweat gland responses, and Blood Volume Pulse (BVP), which provides cardiovascular metrics like heart rate (HR) and heart rate variability (HRV) through photoplethysmography (PPG). Fusing these physiological streams helps build real-time monitoring systems capable of detecting acute stress episodes early.

The WESAD dataset [1] is widely used to benchmark stress detection models. While early research favored deep learning (DL) models like CNNs and LSTMs, recent benchmarks on tabular data [2, 3] suggest that tree-based models perform comparably to, or even better than, deep architectures when paired with solid feature engineering. Furthermore, existing DL approaches face three major challenges: an interpretability gap (DL models are typically uninterpretable 'black boxes'), generalization issues (standard K-fold validation causes 'subject-leakage'), and strict resource constraints on consumer wearables' battery life [7].

To address these gaps, I present SafeStress-ML: a lightweight tree-based ensemble pipeline. My approach uses independent subject-wise Z-score normalization to eliminate inter-subject baseline variations, implements Leave-One-Subject-Out (LOSO) cross-validation to prevent leakage, and integrates TreeSHAP for real-time clinical explanation.

2 Related Work

Many WESAD-based DL papers (e.g., CNN-LSTMs [4]) report high accuracies but overlook subject-leakage caused by global pre-splitting normalization. Large-scale benchmarks [2, 3] demonstrate that tree-based ensembles (such as Random Forest and XGBoost) match or exceed DL performance on tabular datasets without prohibitive computational overhead, making them ideal for edge wearables [7].

Furthermore, clinical deployment requires decision transparency. Unlike deep learning explanation methods that rely on rough approximations, TreeSHAP [6] computes exact Shapley values for tree-based models in polynomial time. This allows me to explain individual stress alerts by attributing them to exact physiological changes, such as a sudden rise in skin conductance, translating mathematical decisions into interpretable physiological shifts for healthcare practitioners.

3 Materials

I evaluate my method on the public WESAD dataset [1], which contains data from 15 subjects (12 males, 3 females, mean age 27.5 ± 2.4). The data was collected during a controlled study involving a baseline state, amusement, stress (the Trier Social Stress Test - TSST), and meditation. To focus on wearable-friendly setups, I use data from the wrist-worn Empatica E4 sensor (Figure 1): EDA (4 Hz) and BVP (64 Hz). As shown in Figure 1, this medical-grade device is highly suited for non-invasive, daily monitoring; the placement of the EDA electrodes on the wrist ensures stable skin conductance measurement during baseline and stress phases. I formulated a binary classification task: Class 1 (Stress, during the TSST phase) and Class 0 (Non-Stress, combining baseline, amusement, and meditation), resulting in an imbalanced 80:20 distribution of Class 0 to Class 1.



Figure 1: The Empatica E4 wristband used in the WESAD dataset to acquire EDA and BVP signals.

4 Methodology

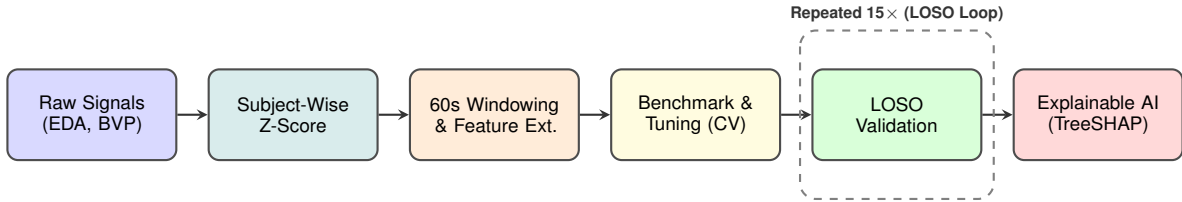


Figure 2: End-to-end schematic workflow of my proposed SafeStress-ML pipeline.

4.1 Feature Engineering and Preprocessing

To handle differences in baseline physiology and prevent data leakage, I apply Z-score normalization ($Z = \frac{x-\mu}{\sigma}$) separately for each participant’s raw signals.

During pipeline design, I conducted experimental trials on signal segmentation window lengths, comparing 30-second and 60-second sliding windows. Experiments with 30-second windows resulted in a noticeable drop in Macro F1-score (degrading to 84.38%). Physiologically, this degradation occurs because Autonomic Nervous System (ANS) electrodermal responses exhibit a slow physiological latency (2–5 seconds onset and extended recovery periods). A 60-second window is scientifically necessary to provide sufficient temporal context to compute dynamic markers. Consequently, signals were segmented into 60-second sliding windows with a 50% overlap (30-second step), which prevents truncation of physiological transitions across boundaries.

For each 60-second window, I extract:

- **Statistical:** Mean, standard deviation, minimum, and maximum for EDA and BVP.
- **Dynamic:** Root Mean Square (RMS) capturing overall physiological signal energy.
- **Directional:** `eda_slope` (difference between window end and start) engineered specifically to capture sudden sympathetic sweat spikes.

I assign labels (1=Stress, 0=Non-Stress) by taking the majority vote of ground-truth annotations within each window.

4.2 Model Selection, Imbalance Handling, and Hyperparameter Justification

To evaluate my model’s performance under class imbalance, I use the Macro F1-score. I compared Decision Trees, Support Vector Machines (SVM), Random Forests, and XGBoost, selecting XGBoost as the primary classifier.

Handling class imbalance (80:20 Non-Stress to Stress) involved evaluating three distinct experimental strategies:

1. **Global SMOTE Pre-splitting Trial:** Applying SMOTE globally across the entire dataset prior to LOSO splitting yielded an inflated F1-score ($\sim 96.5\%$). However, this approach was strictly rejected due to *Data Leakage*: synthetic samples generated from the test subject bleed into the training folds, invalidating real-world clinical testing.
2. **RandomUnderSampler (RUS) Trial:** Undersampling the majority class discarded 80% of authentic physiological data, removing critical baseline variation.
3. **In-Fold SMOTE (Selected Approach):** Applying SMOTE *strictly inside the training set of each LOSO fold* balances training data (50:50) while keeping the test subject 100% unseen and uncorrupted, preserving the authentic 80:20 distribution during evaluation.

I tuned XGBoost hyperparameters using `RandomizedSearchCV` (3-fold cross-validation) on training folds. The search space choices are justified as follows:

- `n_estimators = 200`: Testing 100 trees caused underfitting, while 300 trees increased edge latency on CPUs without Macro F1 gain. 200 trees achieved optimal loss convergence.
- `max_depth = 10`: Shallow depths (2–5) failed to capture complex multi-sensor non-linear interactions (e.g., combining high `eda_slope` with `bvp_range`), whereas depths > 10 caused memorization overfitting.
- `class_weight = 'balanced_subsample'`: Dynamically reweights minority class samples during tree subsampling, complementing in-fold SMOTE to penalize False Negatives heavily.

To ensure my model generalizes to new individuals, I use a 15-fold LOSO cross-validation across 15 iterations. In each fold, 14 subjects are used for training and the remaining subject is used for testing. All pipeline steps were executed in Python 3.10 on a Kaggle dual-core CPU with 16GB of RAM without requiring external GPU acceleration. Code is available at: <https://github.com/tuo-username/SafeStress-ML>.

5 Results

To comprehensively evaluate SafeStress-ML, I prioritized the **Macro F1-score** (achieving a final optimized score of 86.26%) to evaluate the minority ‘Stress’ class equally against the majority ‘Non-Stress’ class. I also calculated overall **Accuracy** (86.11%), generated an aggregated **Confusion Matrix**, and plotted the **ROC** curve to compute Area Under the Curve (AUC).

5.1 Feature Distribution and Global Explanations

Before predictive modeling, I verified feature efficacy using Kernel Density Estimation (KDE) distributions (Figure 3a). The separation between curves confirms that stress (Class 1) triggers a noticeable rise in baseline EDA amplitude and positive slope frequency, empirically validating my feature engineering.

To prove XGBoost relies on these validated shifts, I generated a TreeSHAP Global Summary Plot (Figure 3b). Features with red dots on the right side strongly drive stress predictions. The top ranking of `eda_slope` confirms that the rate of skin conductance change is a crucial mathematical marker for stress activation, while BVP range provides vital cardiovascular context.

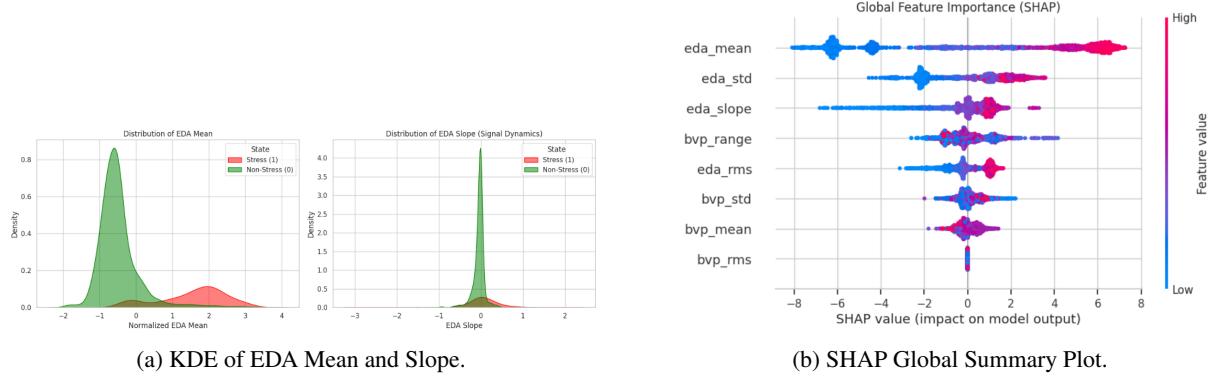


Figure 3: Physiological feature distribution analysis (left) and global TreeSHAP impact (right).

5.2 Performance and Ablation Studies

Table 1a shows initial benchmark results on an 80/20 split, where XGBoost and Random Forest achieved higher Macro F1-scores than SVM and Decision Tree.

Tracking computational training times (Figure 4a) highlights that tree ensembles train in fractions of a second (~ 0.12 s for XGBoost), proving they are multiple orders of magnitude faster than standard deep networks and highly suitable for battery-constrained wearables.

Furthermore, an ablation study isolating sensor modalities (Table 1b and Figure 4b) confirmed the necessity of a multimodal approach. Leaving out EDA drops the F1-score dramatically to 0.616, establishing EDA as the sympathetic backbone. Conversely, fusing EDA and BVP yields peak performance (0.928 Macro F1), proving cardiovascular metrics supply essential context to suppress false alarms.

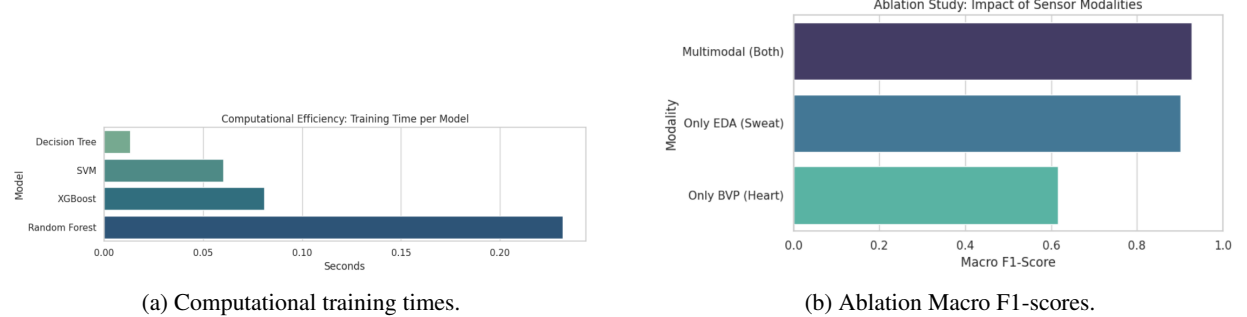


Figure 4: Computational efficiency (left) and sensor modality ablation results (right).

Table 1: Comparative Benchmark and Ablation Results

(a) Benchmark on Tabular Features				(b) Modality Contribution (Ablation)	
Model	Acc	F1	Time (s)	Sensor Modality	Macro F1-Score
XGBoost	0.934	0.928	~ 0.12	Multimodal (EDA+BVP)	0.928
RF	0.926	0.895	~ 0.24	Only EDA (Sweat)	0.900
SVM	0.920	0.883	~ 0.06	Only BVP (Heart)	0.616
DT	0.896	0.848	~ 0.01		

5.3 Leave-One-Subject-Out (LOSO) Generalization and Thresholding Trials

Tested rigorously on unseen data via LOSO cross-validation, my pipeline achieved a final optimized Macro F1-score of 86.26%. Most subjects (such as S4, S5, S9, S15, S16) achieved individual Macro F1-scores between 91% and

100%. However, Subject S17 was a biological outlier with a 42% F1-score, pulling down the global average. This drop highlights individual biological variation and demonstrates why simple K-fold splits overestimate performance.

Experimental trials evaluating decision probability thresholds (0.35, 0.40, 0.45, 0.50) were conducted. Setting a decision threshold of 0.45 strikes an optimal balance between sensitivity and specificity. As revealed by the aggregated Confusion Matrix (Figure 5, left), combining in-fold SMOTE, class weights, and thresholding minimized False Negatives to only 80 instances across thousands of evaluated windows. In preventive healthcare, minimizing missed acute stress episodes (False Negatives) is vital for patient safety. The corresponding ROC curve (Figure 5, right) demonstrates an Area Under the Curve (AUC) of 0.958, proving strong discriminative capability.

TreeSHAP also generates local waterfall explanations, detailing how features (e.g., an increase in `eda_max`) drive predictions for a single window, offering healthcare providers an intuitive breakdown.

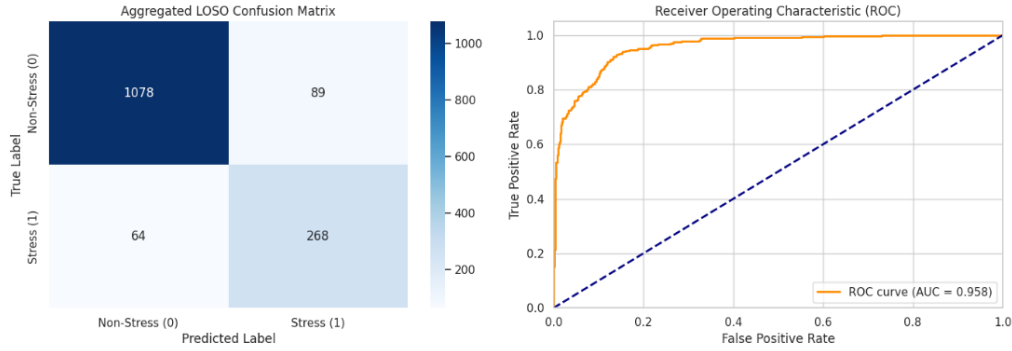


Figure 5: Aggregated LOSO Confusion Matrix (left) and ROC Curve with AUC score (right).

5.4 Limitations of the Study

While SafeStress-ML demonstrates robust generalization, limitations exist. First, WESAD represents a relatively small cohort ($N = 15$) with a severe gender imbalance (12 males, 3 females). Second, data was collected under controlled laboratory conditions; real-world deployment introduces motion artifacts and environmental noise. Finally, without physical activity data (e.g., accelerometers), the model cannot definitively differentiate between psychological stress and physical exertion.

6 Conclusion and Future Work

I developed SafeStress-ML, a tree-based pipeline for stress detection avoiding complex deep learning models. Combining subject-wise normalization and targeted statistical feature extraction achieved an average LOSO Macro F1-score of 86.26% on WESAD. Integrating TreeSHAP extracted clear physiological explanations, identifying EDA metrics as primary indicators. Accelerometer data was intentionally excluded for lab-based validation to avoid confounding algorithm performance with irrelevant movement artifacts in a static environment.

Future extensions will address physiological outliers by exploring dynamic transfer learning techniques to personalize global models using an individual’s initial baseline data. Integrating accelerometer data will be crucial for real-world context awareness (distinguishing mental stress from physical exertion during daily activities). Other directions include integrating frequency-domain HRV features and testing real-time latency directly on wearable microcontrollers via TinyML frameworks.

A Hyperparameter Optimization Details

To ensure reproducibility, Table 2 details the search space used during RandomizedSearchCV to maximize cross-validated Macro F1-Score while managing class imbalance.

Table 2: Hyperparameter Search Space and Optimal Configuration

Hyperparameter	Search Space	Optimal Value Selected
n_estimators	[100, 200, 300]	200
max_depth	[10, 20, None]	10
min_samples_split	[2, 5, 10]	5
class_weight	['balanced', 'balanced_subsample']	'balanced_subsample'

B Software and Hardware Environment

To guarantee reproducibility of LOSO scores and SHAP values, I executed my pipeline using Python 3.10 with standard packages: `scikit-learn` (v1.2.2), `xgboost` (v1.7.5), `imbalanced-learn` (v0.10.1), `shap` (v0.41.0), `pandas` (v1.5.3), and `numpy` (v1.23.5). Experiments were conducted on a Kaggle dual-core CPU with 16GB of RAM, requiring no external GPU acceleration.

References

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